4

Voltage Source to Current Source Conversion

For any practical voltage source, if the ideal voltage is V and the internal resistance is R, the voltage source can be replaced by an equivalent current source I with the internal resistance R in parallel with the current source.

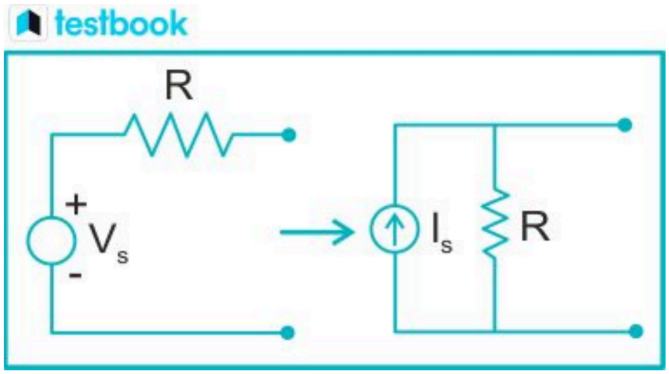


Fig-Voltage Source to Current Source Conversion
Here, Is=VR

5

Steps for Source Transformation

1. **Identify the part of the circuit suitable for transformation.**
2. **Apply the transformation using the above equations.**
3. **Simplify the resulting circuit to analyze it easily.**

Advantages of Source Transformation

The source transformation technique offers several benefits, particularly when dealing with complex circuits:

1. **Simplifies Circuit Analysis: By reducing the number of components in a circuit, source transformation makes the overall analysis faster and more manageable, especially for circuits with multiple sources and resistors.**

2. **Reduces Complexity: It helps in reducing a multi-source circuit into simpler single-source circuits, often making it easier to apply methods like Ohm's law, Kirchhoff's Current Law (KCL), and Kirchhoff's Voltage Law (KVL).**

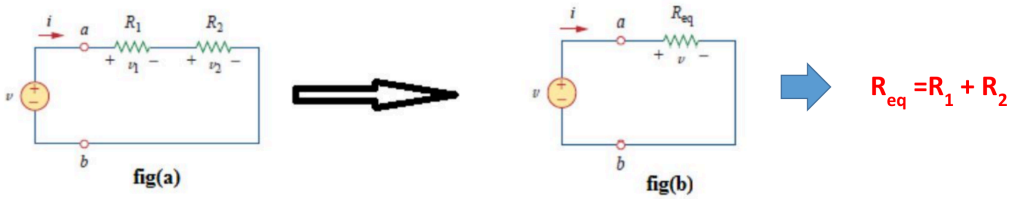
3. **Efficient Use in Theoretical and Practical Circuits: Whether you're dealing with theoretical problems or real-world applications, the ability to transform sources helps in both DC and AC circuit analysis.**

4. **Improves Accuracy in Competitive Exams: In exams, quickly reducing circuits using source transformation techniques ensures faster and more accurate results, particularly in subjects like network theory.**

5. **Flexibility in Complex Circuit Designs: Engineers use source transformation when working with large-scale circuits to reduce their complexity without altering the performance or behaviour of the system.**

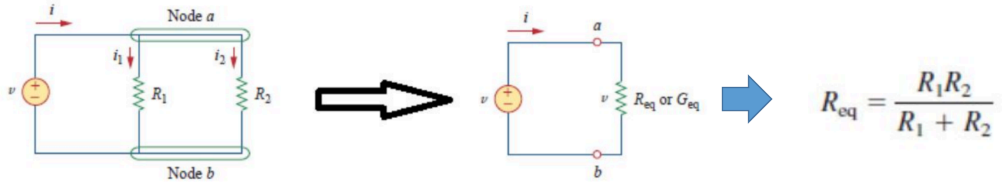
6

Series Connection of Resistors: *Two or more resistors in a circuit are said to be in series when the current flowing through all the resistors is the same.*



fig(a) fig(b)

Parallel Connection of Resistors: *Two or more resistors in a circuit are said to be in Parallel when all the resistors are connected to the same nodes and the same voltage is appearing across all these elements.*



fig(a) fig(b)

10

Example 1

In the circuit at right, find v_{R4}.

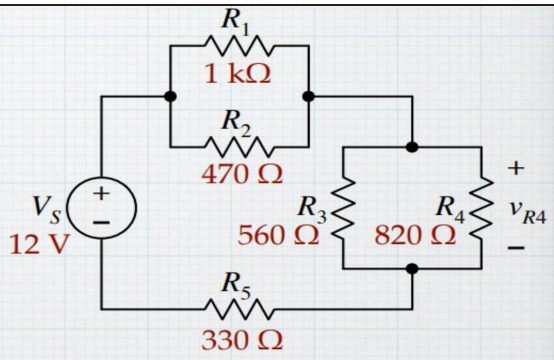
Since R_3 and R_4 are in parallel, they have the same voltage and we can use the parallel equivalent. We can also find the parallel equivalent of R_1 and R_2.

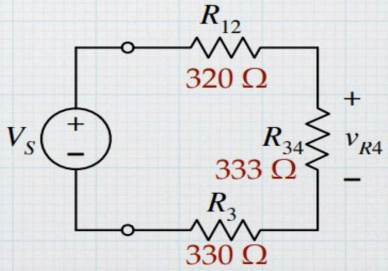
$$R_{34} = \frac{R_3 R_4}{R_3 + R_3} = 333\,\Omega$$

$$R_{12} = \frac{R_1 \cdot R_2}{R_1 + R_2} = 320\,\Omega$$

Then use a voltage divider on the simplified circuit.

$$v_{R4} = \frac{R_{34}}{R_{12} + R_{34} + R_5} \cdot V_S = 4.065\text{ V}$$





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Example 2

In the circuit at right, find i_{R2}.

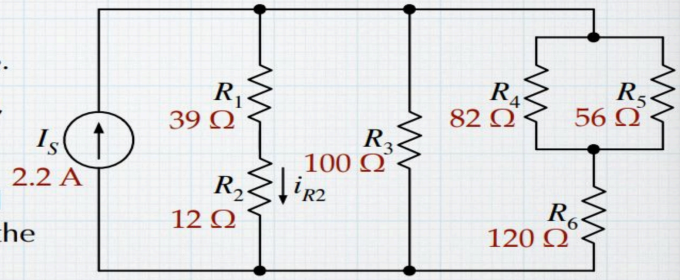
Since R_1 and R_2 are in series, they have the same current, and we can use the series equivalent. We can also find the equivalent resistance of the branch with R_4, R_5, and R_6.

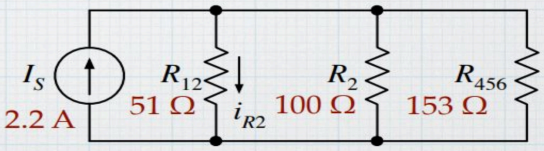
$$R_{12} = R_1 + R_2 = 51\,\Omega$$

$$R_{456} = \frac{R_4 \cdot R_5}{R_4 + R_5} + R_6 = 153\,\Omega$$

Then use a current divider on the simplified circuit.

$$i_{R2} = \frac{\frac{1}{R_{12}}}{\frac{1}{R_{12}} + \frac{1}{R_3} + \frac{1}{R_{456}}} \cdot I_S = 1.19\text{ A}$$





12 Problem : Three resistors are connected together to form a current divider circuit as shown below. If the circuit is fed from a 100 volts supply with a capacity of 1.5kW. Calculate the individual branch currents using the current division rule and find the equivalent circuit resistance.

